

CrewAI Comparison Report

Sequential vs. Hierarchical vs. Single-Agent

1. Objective

This report compares three approaches for the same game-sales analysis task: a CrewAI sequential workflow, a CrewAI hierarchical workflow, and a single-agent baseline. The comparison focuses on output quality, execution time, token usage, approximate cost, reliability, and complexity.

2. Task

The system analyzes game-sales data and produces the top game genres and publishers by Global_Sales, data-driven insights, and a stakeholder-ready summary.

Agent	Responsibility
Sales Data Analyst	Query and extract sales rankings.
Insight Strategist	Analyze extracted results and identify meaningful insights.
Stakeholder Report Writer	Convert findings into an executive-ready summary.

The hierarchical version adds a manager agent that delegates work to specialist agents and reviews their outputs.

3. Execution Results

Approach	Execution Time	Tokens	Approx. Cost	Status
Single-Agent	41.99 s	4,110	\$0.002482	Completed
Sequential Crew	146.63 s	5,671	\$0.003586	Completed
Hierarchical Crew	184.33 s	10,976	\$0.006958	Completed

The hierarchical run used substantially more tokens because the manager coordinated specialist agents and reviewed or delegated their work.

4. Sequential Crew

The sequential workflow executes the specialist tasks in a predefined order: Sales Data Analyst → Insight Strategist → Stakeholder Report Writer. Each downstream task receives the output of the previous task as context.

Results

Genre	Global Sales
Role-Playing	160.51
Platform	89.98
Sports	85.56
Action	56.20
Shooter	54.57

Publisher	Global Sales
Nintendo	333.18
Take-Two Interactive	56.20
Activision Blizzard	36.60
Sony Computer Entertainment	31.19
Bethesda Softworks	26.24

The sequential workflow produced consistent, structured results and maintained clear separation between data extraction, analysis, and reporting.

Advantages: predictable execution order, simple dependencies, lower token usage than hierarchical execution, and easier debugging.

Limitations: the workflow follows a predefined sequence and agents cannot dynamically change the execution plan.

5. Hierarchical Crew

The hierarchical workflow introduces a manager agent that determines which specialist should perform required work, delegates tasks, receives results, and continues the workflow.

The executed run recorded 9 delegation calls through CrewAI's delegation mechanism, demonstrating that the manager actively delegated work rather than simply relying on fixed task-to-agent assignments.

The hierarchical workflow successfully completed the analysis and produced usable results. The final data remained consistent with the sequential workflow, including Role-Playing at 160.51 Global Sales and Nintendo at 333.18 Global Sales.

Advantages: dynamic delegation, manager-based coordination, better support for workflows where task ownership can change, and review/reassignment capability.

Limitations: highest token usage, highest execution time, higher cost, and more complex debugging.

6. Single-Agent Baseline

The single-agent implementation performs the task without multiple specialized agents. It completed successfully in 41.99 seconds using 4,110 tokens at an approximate cost of \$0.002482.

Advantages: fastest execution, lowest token usage, lowest cost, and simplest implementation.

Limitations: one agent handles all responsibilities, role separation is weaker, and complex workflows can become harder to control.

7. Comparison

Factor	Single-Agent	Sequential	Hierarchical
Quality	Good	Good	Good
Execution speed	Best	Moderate	Slowest
Token usage	Lowest	Moderate	Highest
Cost	Lowest	Moderate	Highest
Predictability	High	High	Lower
Specialization	Low	High	High
Dynamic delegation	No	No	Yes
Complexity	Low	Medium	High
Debugging	Easy	Moderate	Harder
Best use case	Simple tasks	Fixed pipelines	Dynamic workflows

8. Cost Analysis

The single-agent approach required approximately 4,110 tokens, while the sequential Crew required 5,671 tokens. The hierarchical Crew required 10,976 tokens, approximately 94% more than sequential and 167% more than single-agent.

The hierarchical run also took approximately 184 seconds, compared with 147 seconds for sequential and 42 seconds for single-agent. The additional cost comes primarily from manager coordination, delegation, and additional agent interactions.

9. Quality Evaluation

Criterion	Result	Observation
Factual Grounding	Strong	Core rankings remained consistent with the dataset/tool results.
Completeness	Strong	Required rankings, insights, and stakeholder summary were produced.
Tone and Usability	Strong	The report-writing stage produced a concise stakeholder-oriented summary.

10. Overall Assessment

For this particular task, the single-agent approach provides the best cost and latency, while the sequential Crew provides the best balance between specialization and execution overhead.

The hierarchical Crew successfully demonstrates genuine manager-driven delegation, but its additional token usage and latency did not produce a clear quality improvement on this relatively structured task.

11. Conclusion

The experiment shows that multi-agent architecture is not automatically better than a single agent. For a predictable three-stage pipeline, sequential execution provides specialization without excessive orchestration overhead. A hierarchical crew is justified when the problem requires dynamic coordination, but its additional complexity and cost should be supported by a corresponding increase in task complexity or output quality.